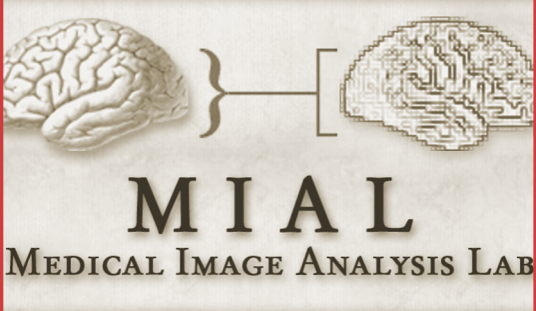


Accounting for Over-Reliance in AI-Assisted Performance Studies



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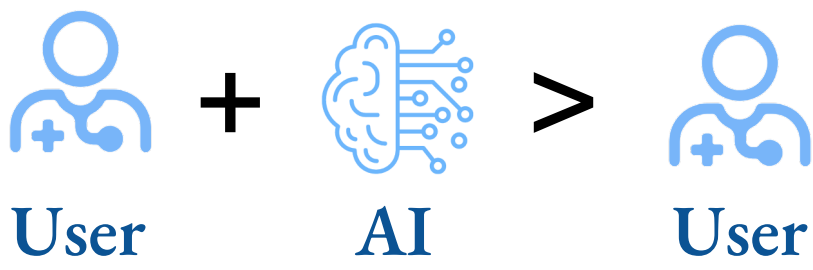
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SFU

Research Motivation To improve the scientific rigor, we conduct a **critical analysis of one aspect in AI-assisted evaluation studies.**

Over-reliance can explain away the performance gain

Is the performance of User+AI better than User alone?



FLAWED

The AI assistance is considered to show a positive effect by helping users improve their performance

Hidden Risks

Over-reliance can lead to long-term consequences, including:

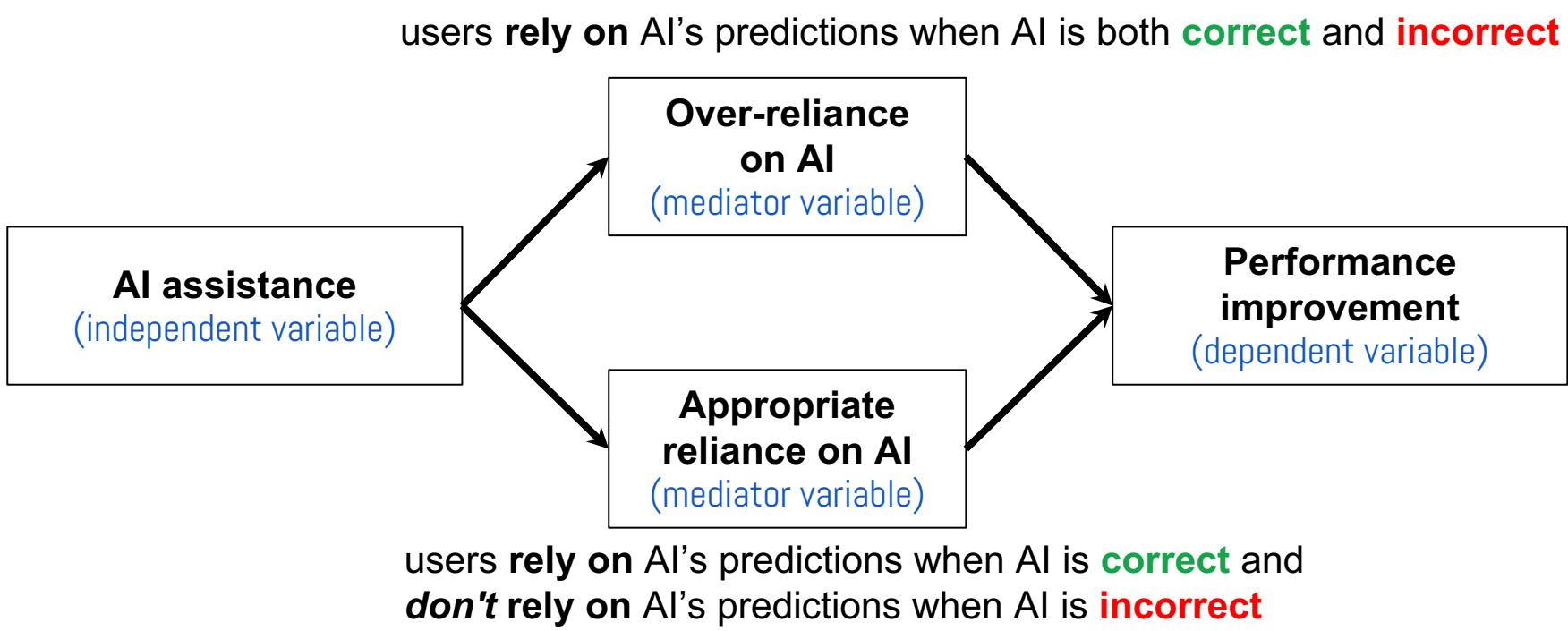
- reduced human autonomy
- professional deskillling

AI-assisted performance gain can be explained by one or two mediators:

- [The upper path] The **negative effect** of over relying on a superior AI assistant regardless of its correctness.

For example, in an exam, if students are equipped with a superior AI assistant, they can simply copy whatever the AI suggests and get higher scores than their own.

- [The bottom path] The **positive effect** of appropriate reliance on AI.



Over-reliance rate can be calculated using a similar fashion as false positive rate

By defining over-reliance rate in this way, its calculation is independent of AI performance and is comparable among different studies.

Confusion matrix for user **reliance**

		User's choice	
		Accept AI's prediction	Reject AI's prediction
AI prediction	Correct prediction	Right acceptance (RA)	Wrong rejection (WR)
	Incorrect prediction	Wrong acceptance (WA)	Right rejection (RR)

$$\text{Appropriate reliance rate (ARR)} = \frac{RA + RR}{RA + RR + WR + WA}$$

$$\text{Under-reliance rate (URR)} = \frac{WR}{WR + RA}$$

$$\text{Over-reliance rate (ORR)} = \frac{WA}{WA + RR}$$

Confusion matrix for **classification**

		Predicted condition	
		Predicted positive	Predicted negative
Actual condition	Real positive	True positive (TP)	False negative (FN)
	Real negative	False positive (FP)	True negative (TN)

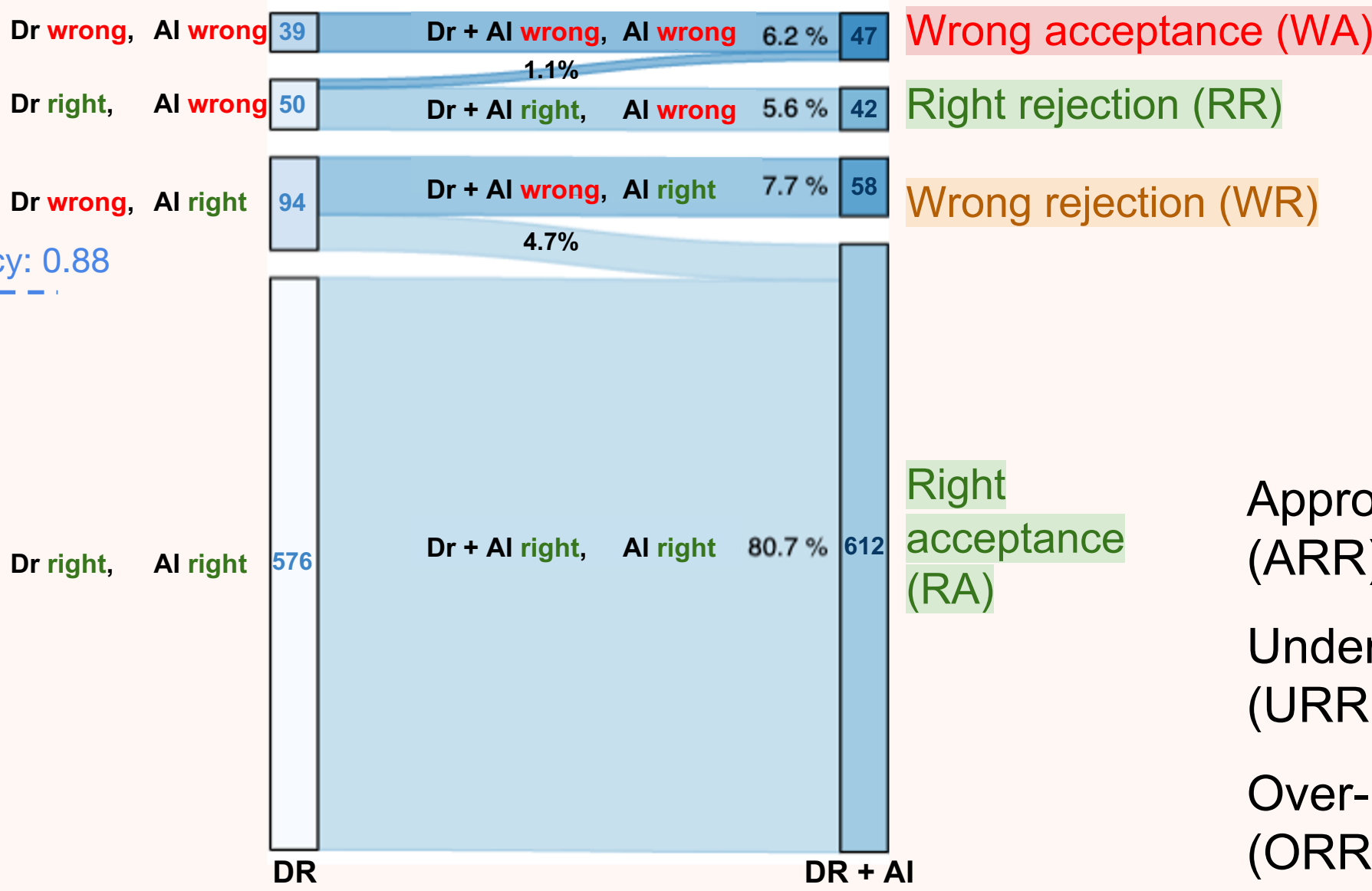
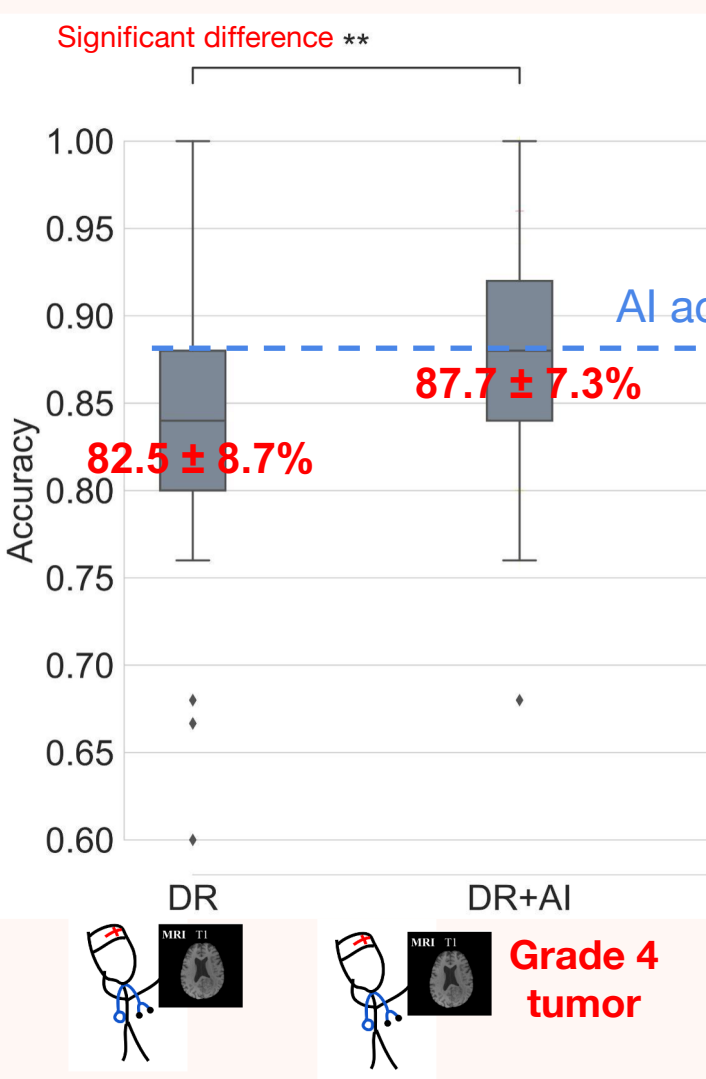
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FN + FP}$$

$$\text{False negative rate (FNR)} = \frac{FN}{FN + TP}$$

$$\text{False positive rate (FPR)} = \frac{FP}{FP + TN}$$

A case study of the phenomenon of over-reliance in an AI-assisted clinical user study

Case study data is from:
Evaluating the clinical utility of artificial intelligence assistance and its explanation on the glioma grading task.
W Jin*, M Fatchi*(first co-author), R Guo, G Hamarneh. AI in Medicine. 2024



Case study result

Both **appropriate reliance** and **over-reliance** explain the performance gain from AI assistance.

	DR baseline	DR + AI
Appropriate reliance rate (ARR) higher is better	0.82 $\frac{576 + 50}{576 + 50 + 94 + 39}$	0.86 $\frac{612 + 42}{612 + 42 + 58 + 47}$ ↑
Under-reliance rate (URR) lower is better	0.14 $\frac{94}{94 + 576}$	0.09 $\frac{58}{58 + 612}$ ↓
Over-reliance rate (ORR) lower is better	0.44 $\frac{39}{39 + 50}$	0.53 $\frac{47}{47 + 42}$ ↑

Suggested changes in human-AI collaboration studies to account for over-reliance

1. Using complementary performance, instead of performance gain, as the study endpoint.

- Studies using performance gain as the endpoint without addressing the influence of over-reliance can mislead readers and the public to default the performance gain to AI-assisted appropriate reliance.
- To avoid the pitfalls of performance gain being the study endpoint, consider replacing the endpoint with complementary performance:
user+AI > max(user, AI), which is a strong indicator of appropriate reliance.

2. Controlling, measuring, and reporting over-reliance.

Since over-reliance may not be completely eliminated but can be mitigated, it is the responsibility of the researchers to control, reduce, measure, and report over-reliance in human-AI collaboration study.

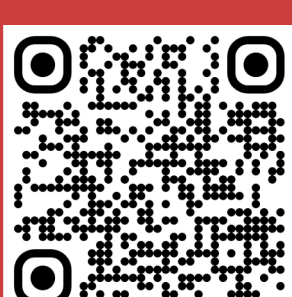
- AI system design and study design:** encourage users' skepticism by being able to reliably indicate the limitations, scope, weaknesses, uncertainty, and evidence for AI's suggestions.
- Study measure and reporting:** collect data on participants' agreement or disagreement with AI suggestions, measure and report over-reliance rate.

1. Disclosing and discussing the influence of over-reliance when interpreting study results.

- If performance gain is used as the study endpoint, the discussion should explicitly discuss the role of over-reliance in explaining away the mechanism of performance gain, and to disclose that the performance gain cannot be totally attributed to the positive aspects of using AI, but also includes the negative effect of using AI of over-reliance.



Paper and Project Webpage
weina.me/overreliance



Video Presentation of the Paper
<https://youtu.be/bIUyLrWFlXc>